



TM280

Condition monitoring based on vibration measurement

Requirements

Training modules	B&R product range TM210 – Working with Automation Studio
Software	Automation Studio 4.2.5 and later Automation Runtime 4.25 and later mapp V1.40 and later
Hardware	ETA280.0102-100

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1 Introduction

From oscilloscope modules to sensor-equipped servo drives, from the monitoring chip inside every Automation PC to vibration analysis modules with embedded intelligence, B&R offers every possibility to keep your finger on the pulse of the machine without requiring any external units. Condition monitoring is already an integral part of every B&R system and consequently of every machine that is automated with B&R technology. Predictive maintenance based on constant condition monitoring helps avoid unplanned downtime and reduces the cost of stocking spare parts. Using a machine's status signals as an input for open and closed-loop control systems helps extend its service life.



Figure 1: Condition monitoring is used in a wide variety of industries

The development efforts of mechanical engineers in machine manufacturing are paying off as the power density and functional density of production machines continue to increase. As a result, fewer machines are needed to produce the same quantity and level of complexity. More efficient, economical production makes machine operators more competitive. On the other hand, this increased efficiency also makes them more dependent on each individual machine. The more units that can be produced per hour, the more each hour of downtime costs. And anyone who has ever operated a machine knows that downtime can never be completely avoided.

1.1 Learning objectives

This training module takes a closer look at condition monitoring based on vibration measurement.

- Participants will receive general information about the topic of condition monitoring
- Participants will learn the basics of vibration measurement technology.
- Participants will learn about the areas of application and use cases.
- Participants will learn about the condition monitoring module - X20CM4810.
- Participants will receive information about practical applications for damage detection.
- Participants will receive information about condition monitoring in a real machine environment.

2 General information about condition monitoring

When managers are considering whether or not to invest in new machinery, they evaluate not only the purchasing price, but the cumulative cost over its entire useful life. In order to reduce these overall costs, often referred to as TCO (total cost of ownership), the goal is to maximize machine availability while minimizing the time and money spent on maintenance.

One of the ways machine manufacturers are attempting to optimize maintenance costs is to replace fixed maintenance intervals with condition-based preventative maintenance. An advantage of this is that maintenance work can be delayed until it is actually needed, which can be quite a bit longer than conservatively estimated maintenance intervals. In addition, scheduling freedom is retained and maintenance work can be performed during regularly scheduled downtime, such as on weekends. At the same time, this solution avoids risking system failure due to neglected maintenance. Success in this balancing act relies on condition monitoring – permanently monitoring conditions throughout the entire system. Data gathered through condition monitoring can be used to identify the maintenance requirements of the respective equipment.

Request	Solution
Machine downtime = costly	Machinery health monitoring
Maintenance = costly	Maintenance control
Poor processing output (quality/quantity) = costly	Process optimization

2.1 Effects on machine condition

There are different influences that can have an effect on the state of a machine.

An important factor here is, for example, the environmental conditions where the work is being done. A very hot environment with relatively high humidity can have an effect. This is also the case if it is very cold or the environment is dusty.

An additional factor here can also be the person. If, for example, issues are not solved professionally or maintenance work is not done properly, this can lead to problems. In addition, incorrect operation by users themselves can of course worsen the state of the machine.

Continued operation of a machine can, in these situations, cause components to wear out.



Eventually machine/system service is necessary.

2.2 Maintenance strategy

The operating performance of every mechanical component changes over the course of operation, with each component becoming defective at some point. It is crucial to recognize such a change before the component can no longer fulfill its function.

For a failure-oriented operational mode ("reactive maintenance"), components are only replaced if they can no longer fulfill their function. For planned operation ("preventive maintenance"), components are replaced at a certain point in time – regardless of their current condition.

For condition-based maintenance, the area where maintenance is planned can be significantly isolated while reducing the risk of failure at the same time.

	Advantages	Disadvantages
Reactive maintenance	- Utilization of reserves with regard to wear - No costs during the period of use	- Unexpected failure - Consequential damage - High downtime costs - Low operational safety
Preventive maintenance	Can be planned well	- No utilization of reserves with regard to wear - Increased risk of failure after maintenance - Fixed costs
Condition-based maintenance	- Early recognition of problems - Downtime can be planned - Utilization of reserves with regard to wear - High operational safety - Avoidance of consequential damage	- Dealing with the issue - Investment costs

Table 1: Maintenance strategies

2.3 Condition monitoring – How?

Information must be collected in order to be able to monitor the state of the machine, and from that information corresponding conclusions have to be drawn regarding the required maintenance of monitored parts – this is referred to as condition monitoring.

This information can be collected by evaluating various signals and condition parameters.

Some significant physical condition parameters are:

- Vibrations
- Velocity
- Motor current
- Temperature
- Pressure
- Flow rate
- Fill level
- Conductivity



Figure 2: Physical measured values



In this training module, we're focusing on condition monitoring based on vibration measurement technology.



Different modules (e.g. analog input or temperature modules) for collecting condition parameters are available in the B&R product portfolio. There are also specialized modules such as valve control modules with integrated switching time detection or oscilloscope function.

3 Condition monitoring based on vibration measurement

In addition to productive output, machines also generate undesired vibrations. These undesired vibrations can be monitored with condition monitoring.

Vibration measurement technology can be used to collect data related to mechanical vibrations that occur on a machine. For this, corresponding vibration sensors (acceleration sensors) are used. The acceleration measured within the scope of the condition monitoring is typically measured with piezoelectric sensors.

3.1 Basics of instrumentation and mathematics

For the most part, structure-borne sound, i.e. the sound that spreads through a solid object, is measured. The three vibration magnitudes correlate mathematically to the integration or differentiation of the basic variables. Vibration velocity is calculated through integration from the vibration acceleration; vibration displacement is calculated through integration from the vibration velocity.

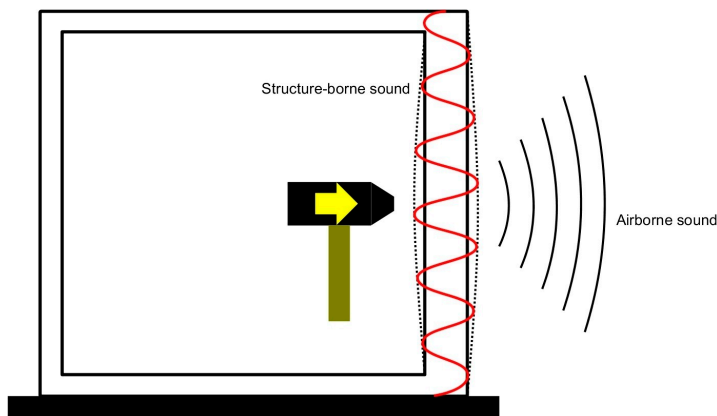


Figure 3: Structure-borne sound

If a fixed medium is stimulated by an impact, structure-borne sound spreads throughout it. This consists of additional frequencies that are determined by the shape of the structure and the material it is made from (e.g. gong or concrete block).

3.1.1 Vibrations

Vibrations are forms of movement that occur very frequently in nature. A vibration is a cyclic, i.e. repetitive simultaneous movement of a structure in its rest or equilibrium position.

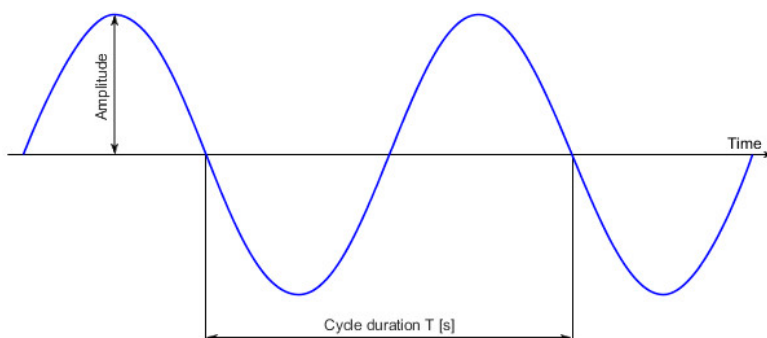


Figure 4: Vibrations

Vibrations have the following parameters, among others:

- Amplitude
- Period duration
- Frequency

Causes and effects

There are numerous overlapping causes for vibrations. The amplitude of the vibration depends on several factors such as attenuation through joints or grease, the rigidity of the component, the housing and foundations, and much more. A few typical causes are explained below.

Imbalance

Imbalances on a rotating structure not only cause forces on the bearing and foundations, but also vibrations in the machine.

The character of these vibrations is harmonic. The excitation frequency corresponding to the rotary frequency of the imbalanced rotor.

Impacts

Foreign objects as well as loose or colliding parts can cause shocks between rotating and stationary parts. These shocks repeat periodically once or several times each time the shaft revolves.

The frequency of these shock repetitions corresponds to the rotary frequency of the shaft or its harmonic frequency.

Roller bearing damage

Most bearing damage results from changes on the surface (pitting). By rolling over the damaged area on the inner ring, outer ring, cage or rolling element, pulse-shaped shocks occur that make the bearing structure and its components vibrate.

Each of these shocks appears in the vibration signal through the typical course of a shock sequence. Characteristic values can be obtained from these measurements that give an indication of the condition of the bearing.



The excitation frequency on the inner ring, outer ring, cage and roller bearing damage is specified by the bearing manufacturer.

Effects

Increased vibrations can result in malfunctions in the machine, particularly in measurement and control devices. If this causes the measuring equipment to resonate as well, incorrect measurements will result and manufacturing quality will suffer.

In addition, stress will develop on the components of the machine. Unwanted vibrations result in increased wear with partly plastic distortion of components and increased crack formation all the way up to failure.

3.1.2 Fast Fourier transforms (FFT)

Vibration signals generally consist of a number of vibrations that occur simultaneously and overlap. Individual frequencies are not directly evident from a timing diagram.

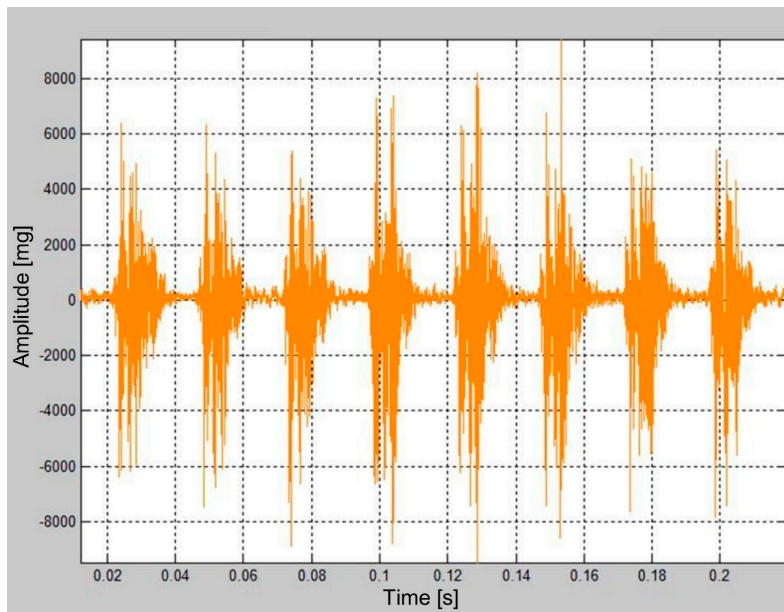


Figure 5: Time signal of a vibration

The Fourier transform is the basic principle of frequency analysis. It assumes that each harmonic vibration can be broken down into any number of sinusoidal and cosinusoidal waves, the sum of which reproduces the original vibration. In order to be able to evaluate individual partial vibrations as amplitude and frequency, the digitized time signal is converted into a frequency spectrum. In addition, a small extract is taken from the signal; this is known as the time window. Using the FFT (Fast Fourier transformation) algorithm, the frequency spectrum is calculated from this so that each involved vibration and its associated frequencies and amplitudes is shown as a single line in the line spectrum.

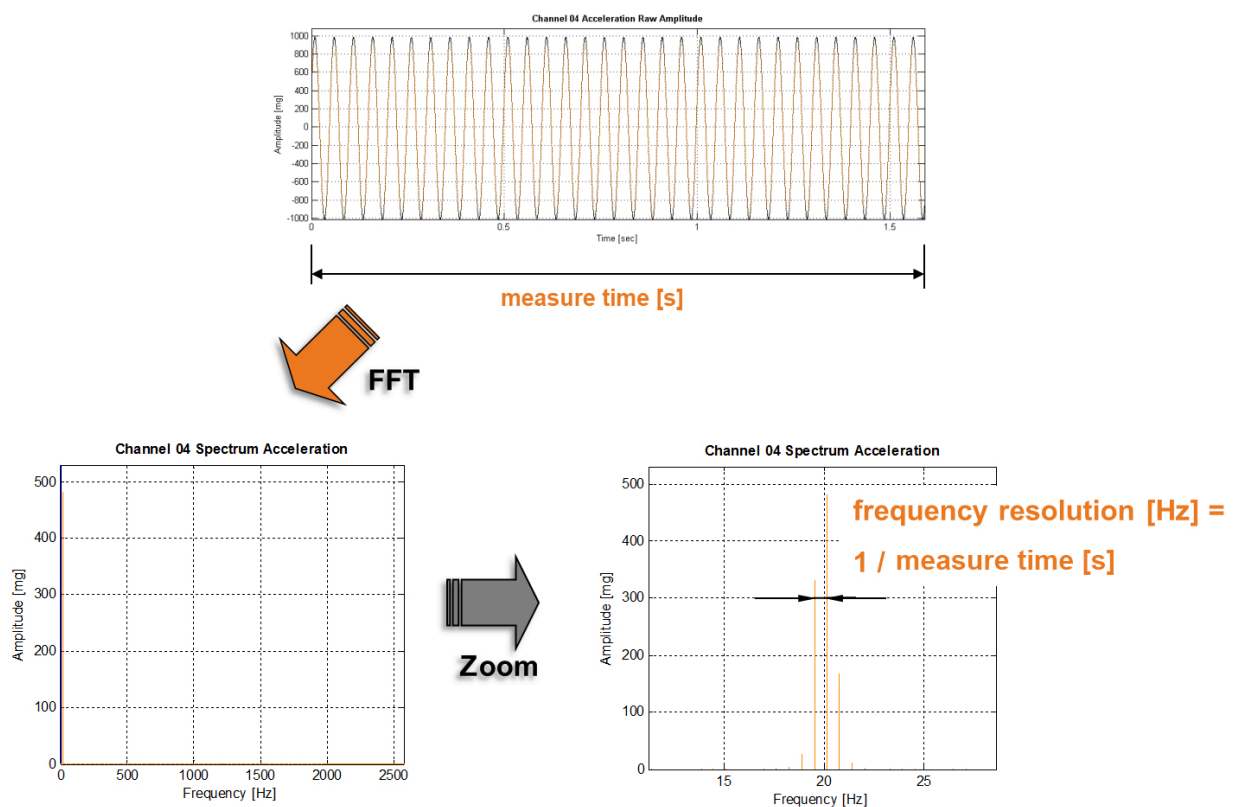


Figure 6: Fast Fourier transformation

A closer look at the spectral lines lets you determine the frequency resolution.
 Frequency resolution [Hz] = $1 / \text{measurement time [s]}$



For a single sine signal with a constant frequency, a single line is shown in the frequency spectrum.

3.1.3 Envelope analysis

The envelope analysis is a method for differentiating between harmonic causes (imbalance, orientation) and impact-related causes (roller bearing damage, gearing damage, etc.).

Here, the frequency spectrum of the envelope signal is evaluated.

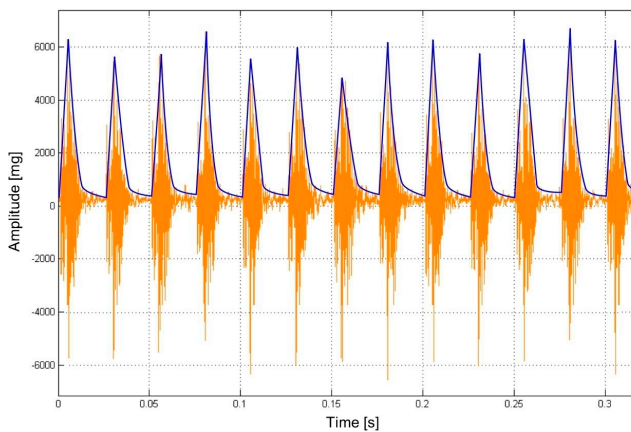


Figure 7: Time signal with envelope (blue)

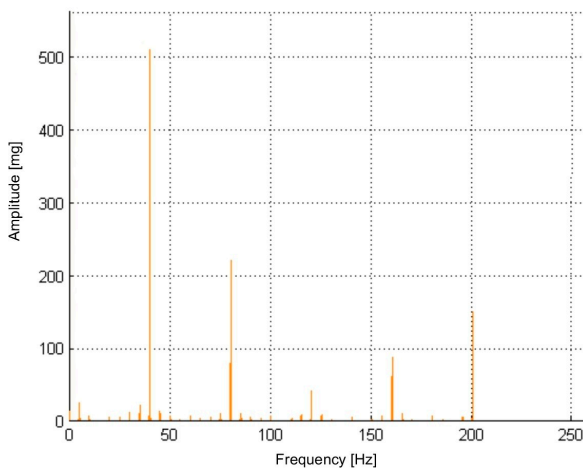


Figure 8: Frequency spectrum of the envelope



Harmonic causes cannot be properly identified in an envelope spectrum.

3.1.4 Sensor technology

Vibration sensors convert the mechanical vibrations of the machines being measured into an electrical signal.

The acceleration measured within the scope of the condition monitoring is typically measured with piezoelectric sensors.

The Integrated Electronics Piezo Electric (IEPE) technology used in B&R sensors strengthens the signal directly in the sensor and emits it as a low-resistance voltage signal. Sensor sensitivity is specified in mV/g.



Piezoelectric sensors cannot measure static magnitudes.

Accelerometers

- IEPE interface
- Sensitivity: 100 mV / g
- Measurement range: ± 50 g
- 2 mounting options



Figure 9: B&R acceleration sensor

3.1.5 Sensor positioning

To ensure optimal detection and measurement of frequencies propagating from a point of damage, sensor positioning is very important. The ideal position for mounting a sensor on a structure is often difficult to reach and is not always necessary.

Since sound waves propagate throughout the entire structure, the damage frequencies are measured with varying intensity or amplitude (green arrow).

If a flexible connection is used, a valid measurement is no longer possible (red arrow).

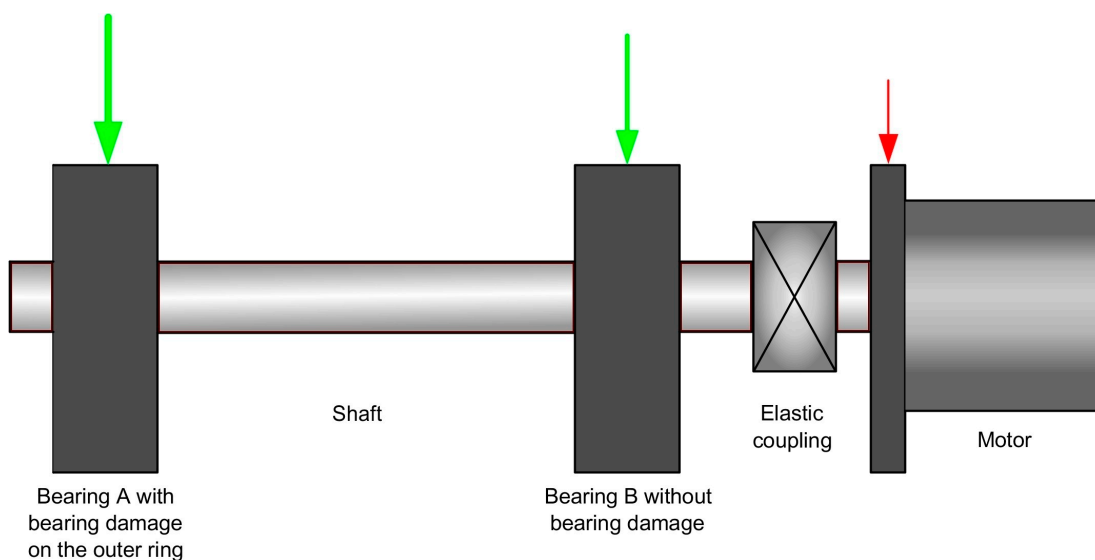


Figure 10: Sensor position



Detailed information regarding the installation of sensors can be found in the data sheet of the associated module.

3.1.6 Determining limits and alarm limits

Some manufacturers give limits for permissible vibrations and other relevant factors for assessing the status of the machine.

For certain machines and systems, limits are fixed by norms. With the exception of ISO 10816, these give little information for assessing the actual status of the machine.

Characteristic values are calculated from the measured signals, which are representative of the status of the system at the given measurement point. A fundamental statement can be made by comparing the characteristic values with predefined limit values.

Operators can also draw on their own experience when assessing the status of the machine. Long-term observation of the characteristic values and the machine's history can provide relevant values based on experience. The characteristic curve of the characteristic values can also be used for state assessment. Changes in the trend are observed with the normal condition as the starting point.

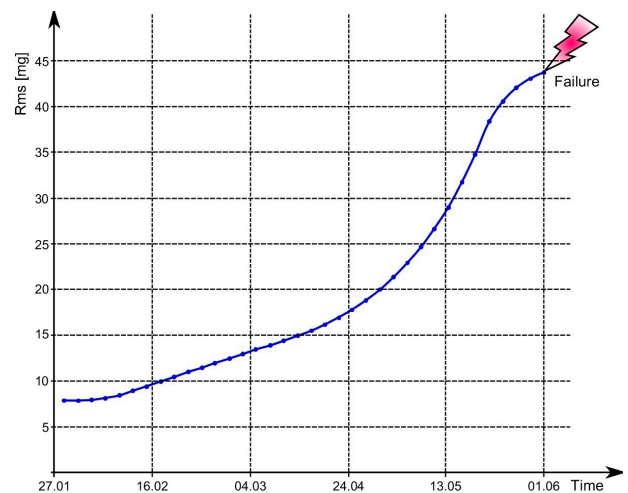


Figure 11: Trend progression

3.2 General parameters and characteristic values

It is possible to gain good insight into the condition of a machine or system by collecting the parameters associated with errors. This parameter data is used by different algorithms to calculate the characteristic values.

Selecting suitable characteristic values and assessing them over a longer period of time is the basis for effective and successful monitoring of a machine.

3.2.1 RMS value

The RMS value is also known as the quadratic mean, or the root-mean-square. Along with the amplitude, it also takes the energy content of the vibration into account and is the mathematical background for many characteristic values of assessment.

The RMS value can be calculated for the raw signal and also for the envelope signal.

ISO 10816

Formed from the raw signal of the vibration velocity in a frequency range from 10 Hz to 1 kHz according to ISO 10816.

VIBRATION SEVERITY PER ISO 10816						
Machine			Class I small machines	Class II medium machines	Class III large rigid foundation	Class IV large soft foundation
in/s mm/s						
Vibration Velocity Vrms	0.01	0.28				
	0.02	0.45				
	0.03	0.71			good	
	0.04	1.12				
	0.07	1.80				
	0.11	2.80		satisfactory		
	0.18	4.50				
	0.28	7.10		unsatisfactory		
	0.44	11.2				
	0.70	18.0				
	0.71	28.0		unacceptable		
	1.10	45.0				

Figure 12: Assessment in accordance with ISO 10816

3.2.2 Peak value

The peak value of a mechanical vibration signal indicate the maximum sum of individual impacts that come from the ambient noise. Different types of damage give rise to strong impacts, which show up in the peak value.

3.2.3 Crest factor

The crest factor is defined as the quotient derived from the peak value and the RMS value. It can be used for detecting bearing damage or poor lubrication.

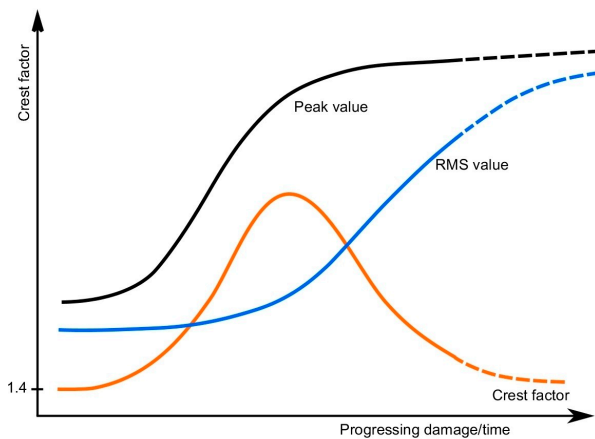


Figure 13: Crest factor



The crest factor can remain unchanged or even sink again, even despite damage.

3.2.4 K(t) value

The $K(t)$ is described in the VDI 3832 guideline and is calculated from the RMS value and the peak value of a time signal for vibration acceleration.

This ratio correlates to reference values that are held shortly after the running-in time.

The $K(t)$ value decreases with progressive wear and can be divided into three classes:

- Undamaged
- Early damage
- Pronounced damage

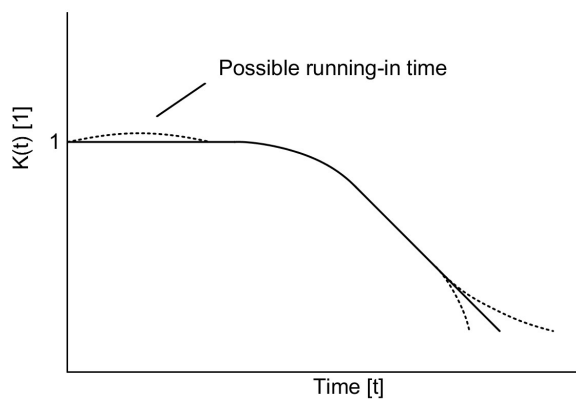


Figure 14: $K(t)$ value progression

3.2.5 Kurtosis

The kurtosis factor is a type of weighted crest factor. This is an effective characteristic value for assessing the number of peaks (steepness, concavity) in a signal. The value is around zero for a normal distribution. A positive value shows a more leptokurtic distribution and a negative value a more platykurtic distribution. It can be used for detecting bearing damage or poor lubrication.

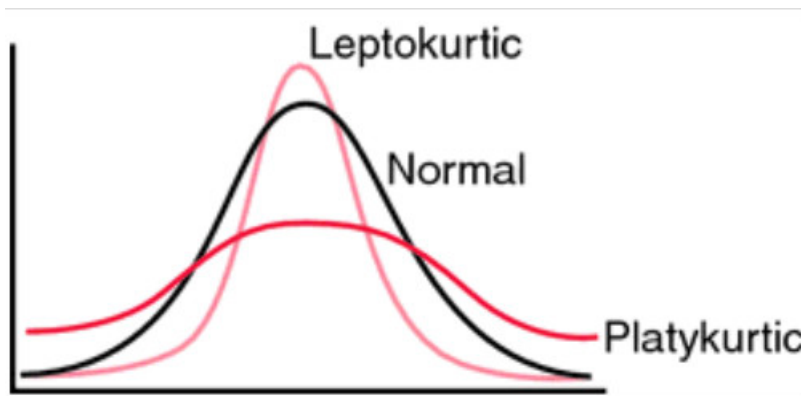


Figure 15: Kurtosis

3.3 Area of use and application examples

The areas of application are manifold, but usually in areas where very high failure or downtime costs arise. An additional criterion is machines that require expensive components, therefore resulting in high service costs. This area of application isn't restricted to monitoring a machine or system, but can also be used for process optimization or improvement.



The goal with condition monitoring is to increase machine availability, lengthen maintenance cycles and also raise the process output (quality and quantity).



Additional information can be found in the data sheet of the associated module.

3.3.1 Wood industry

Application

Wood drying

Goals

Detect bearing damage

Detect accumulation of dust

Benefits

Prevent production downtime

Reducing the maintenance costs

Prevention of fires caused by ignition of sawdust on hot bearings

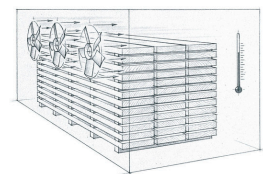


Figure 16: Wood industry

3.3.2 Packaging industry

Application

Label printing machine

Goals

Detect bearing damage

Extension of maintenance intervals

Benefits

Reduce unplanned downtime

Ensure correct contact pressure setting

Ensure optimal print quality (resonance detection)

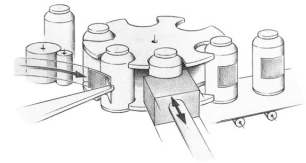


Figure 17: Packaging industry

3.3.3 Textile industry

Application

Industrial washing machines

Goals

Early detection of imbalance

Monitor damage to bearings and suspension system

Benefits

Competitive advantage through faster wash cycle

Save energy and reduce wear on machine

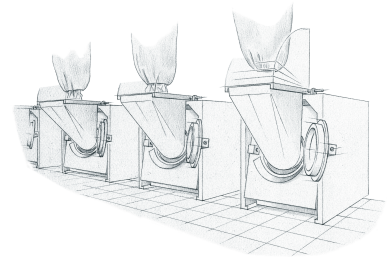


Figure 18: Textile industry

3.3.4 Metalworking industry

Application

Milling machine

Goals

Detect bearing damage

Detecting the amount of mechanical wear

Resonance point monitoring

Benefits

Prevent production downtime

Reducing the maintenance costs

Consistent production quality

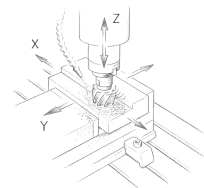


Figure 19: Metalworking industry

3.3.5 Plastics industry

Application

Blow molding machines

Goals

Monitor tool closing process

Monitor machine lubrication

Benefits

Competitive advantage through improved processing quality

Machine availability increased

Reduced maintenance cycles

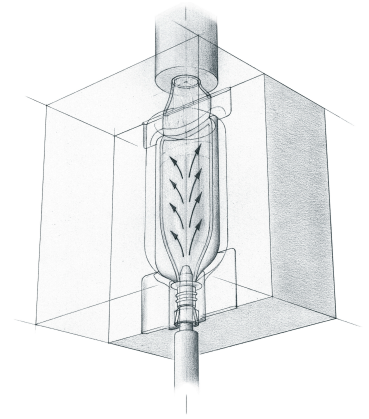


Figure 20: Plastics industry

4 B&R vibration measurement systems

New options result from the integration of vibration measurement systems into the automation system. By setting basic limit values, warnings or an alarm can be issued or even logical correlations with other parameters can be used, such as the load, speed or shape of the trend curve.

4.1 Condition monitoring module - X20CM4810

The X20CM4810 was developed specifically for condition monitoring based on vibration measurement. It is based on integrated data analysis (FFT and envelope analysis) of the input signal and provides different parameters and characteristic values.



Figure 21: Condition monitoring by B&R

Among other things, it can be used to measure imbalance, bearing damage, misalignment or periodically occurring impact on a machine. Acceleration sensors are used to measure vibrations. In addition, the module converts the measured acceleration values into speed values.



With the resolution provided by module, evaluating the acceleration values provides sufficient information, even in the lower frequency domains. B&R therefore recommends using only the acceleration values. For this reason, the calculation of the speed values is disabled by default.



Additional details and information can be found in the data sheet of the module.

4.1.1 Signal processing

The module supports wanted signals up to 10 kHz and uses a maximum sampling rate of 51.5625 kHz. As previously mentioned, it has an integrated data analysis (based on FFT and envelope analysis).

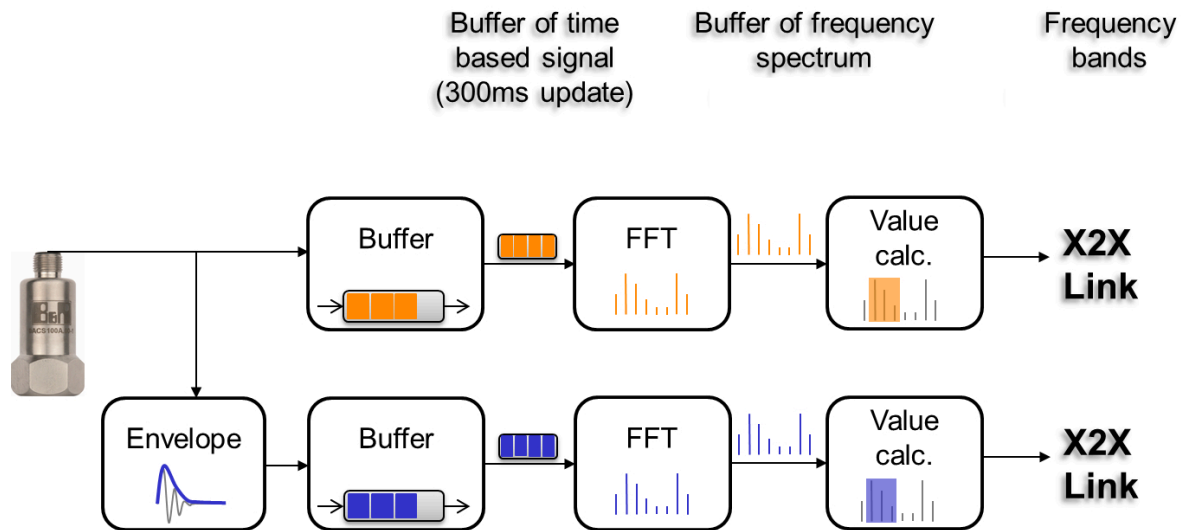


Figure 22: Signal processing block diagram

4.1.2 Filter configuration

The module has a number of configurable filters.

High-pass filter

There is an adjustable high-pass filter for the entire module.

Possible settings are:

- 500 Hz
- 1 kHz
- 2 kHz

Low-pass filter for raw signal

For each channel, there is an adjustable low-pass filter for the raw signal that can be used to control the sampling frequency. Reducing the maximum frequency allows the frequency resolution in the spectrum to be increased. In addition, the lowest frequency of the raw signal that still has to be evaluated is usually defined automatically.

Maximum frequency	Frequency resolution in spectrum	Minimum frequency
10000 Hz	3.1471 Hz	9.441 Hz
5000 Hz	1.5736 Hz	4.720 Hz
2000 Hz	0.6294 Hz	1.888 Hz
1000 Hz	0.3147 Hz	0.944 Hz
500 Hz	0.1574 Hz	0.472 Hz
100 Hz	0.0629 Hz	0.188 Hz

Low-pass filter for envelope signal

For each channel there is an adjustable low-pass filter for the envelope signal. Reducing the maximum frequency allows the frequency resolution in the spectrum to be increased. In addition, the lowest frequency of the envelope signal that still has to be evaluated is usually defined automatically.

Maximum frequency	Frequency resolution in spectrum	Minimum frequency
2000 Hz	0.6294 Hz	1.888 Hz
1000 Hz	0.3147 Hz	0.944 Hz
500 Hz	0.1574 Hz	0.472 Hz
200 Hz	0.0629 Hz	0.188 Hz

4.1.3 Buffering the time signal

The values that have been sampled are stored in the module's internal buffer. The size of the buffer is constant and can store 8192 measured values. This results in the ratio between the sampling frequency and the duration of measurement.

Duration of measurement = Buffer size / Sampling frequency

Since the values stored are dependent on the configured sampling frequency and not the hardware-based sampling frequency, not all values that are measured are stored. At a measurement duration of 318 ms, every second value is stored; at a duration of 15.9 ms, every hundredth value is stored.

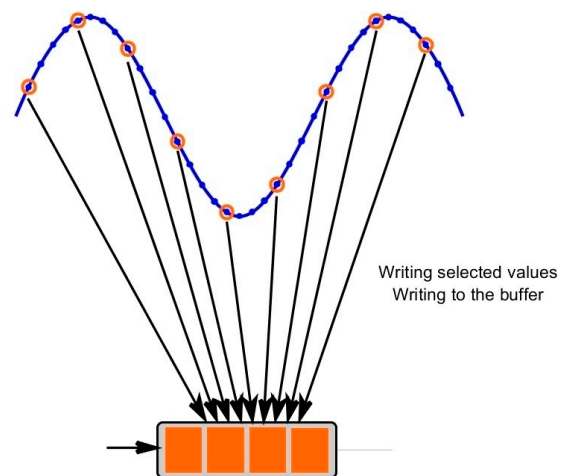


Figure 23: Ring buffer

Maximum frequency	Sampling frequency	Measurement duration
10000 Hz	25781 Hz	0.3178 s
5000 Hz	12891 Hz	0.6355 s
2000 Hz	5156 Hz	1.5888 s
1000 Hz	2578 Hz	3.1775 s
500 Hz	1289 Hz	6.3550 s
200 Hz	516 Hz	15.8875 s

Table 2: Overview of possible measurement times



Every 300 ms, a copy from the buffer is forwarded for processing, i.e. every 300 ms the parameters and characteristic values are updated.
There is also the possibility to upload the module-internal buffer of the module.

4.1.4 Parameters and characteristic values

The module automatically calculates parameters and characteristic values from the ring buffer via integrated data analysis and they are updated every 300 ms. How current the data is for the analysis can be influenced via the sampling frequency (or low-pass filter for raw signal).

For the calculation, the following sources are differentiated between:

- Raw signal
- High-pass filter raw signal (high frequency)
- Envelope signal



Hardware \ X20 system \ X20 modules \ Other modules \ X20CM4810 \ Register description \ Values

Raw signal

This is the raw signal for vibration acceleration or speed between the configured minimum and the maximum frequency (low-pass filter raw signal).

The following parameters and characteristic values are determined for this area:

- RMS value
- Peak value
- Crest factor
- K(t) value
- Kurtosis

High-pass filter raw signal (high frequency)

This is the high-pass filter raw signal for vibration acceleration in the frequency domain between the set high-pass filter value and 10 kHz.

The following parameters and characteristic values are determined for this area:

- RMS value
- Peak value
- Crest factor

Envelope signal

This is the envelope signal of the vibration acceleration or speed between the configured minimum up to the maximum frequency (low-pass filter envelope signal).

The following parameters and characteristic values are determined for this area:

- RMS value

4.1.4.1 Exercise – Characteristic values and parameters

While doing the exercise, the training structure is moved and the impact on the parameters observed.

Exercise: Characteristic values and parameters

- 1) Create a new Automation Studio project
- 2) Add training assembly hardware
- 3) Switch on the IEPE sensor supply for the channels via the I/O configuration

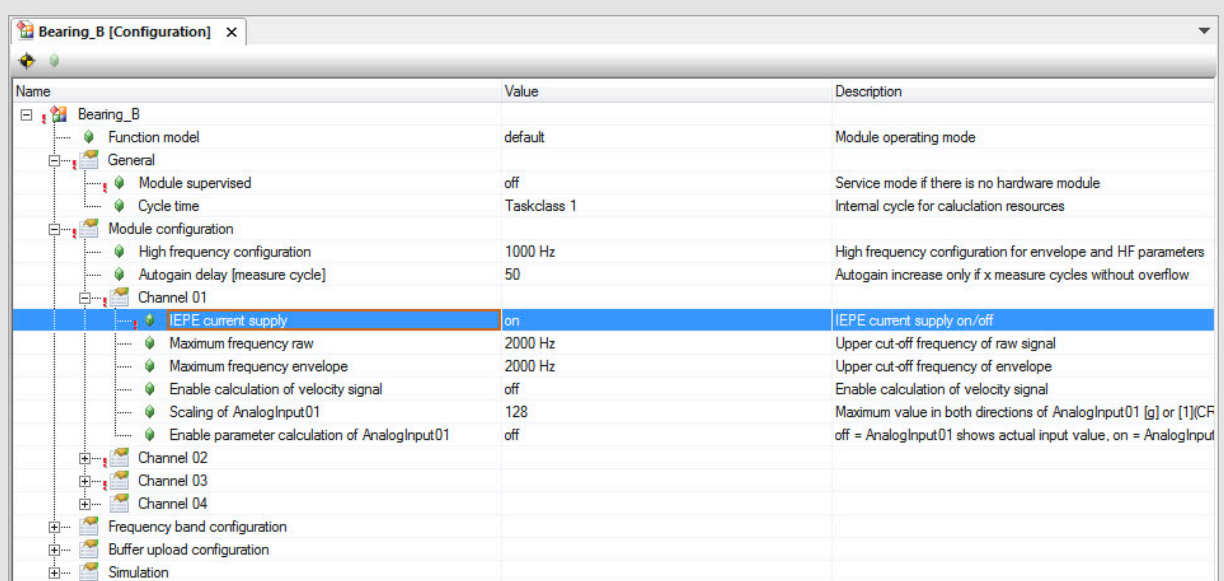


Figure 24: Activate IEPE sensor supply



The sensor needs to be supplied with power in order to take measurements. The option of turning the power supply on and off exists in order to allow a sensor to be connected to two different channels. One channel supplies the sensor with power (IEPE current supply on), so the supply should be turned off for the second channel (IEPE current supply off).

- 4) Offline installation or project installation and startup of the training structure
- 5) Open the I/O mapping of the condition monitoring module
- 6) Observe characteristic values
- 7) Move the training structure by hand, deal an impact and observe characteristic values

4.1.5 FFT and frequency bands

An FFT algorithm converts the measurements in the ring buffer into a frequency spectrum with spectral lines. The frequency resolution should be set as high as possible to obtain the most precise results possible.

Frequency resolution [Hz] = 1 / measurement time [s]



8192 measurements in the time domain produce 4096 (8192 / 2) spectral lines.

Restrictions can be made in the frequency domain using frequency bands in order to be able to better identify certain fault events.

Different damage frequencies can be monitored in parallel by simultaneously using several frequency bands.

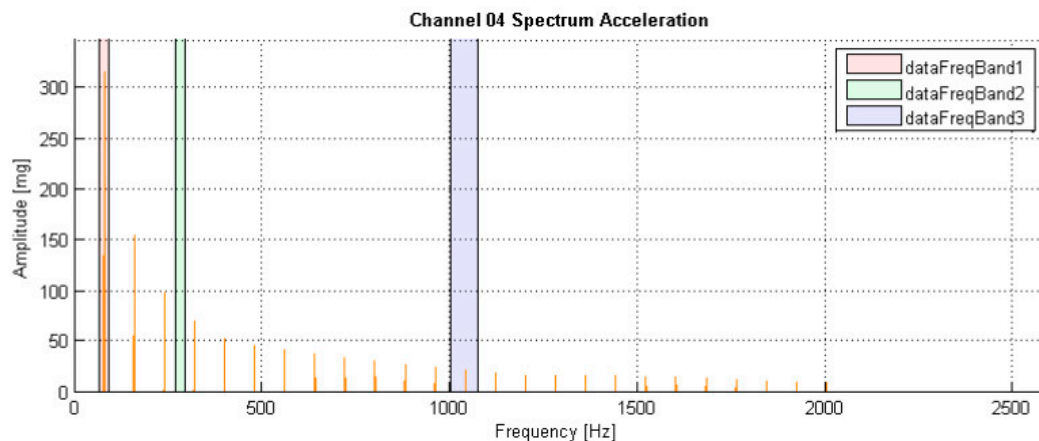


Figure 25: Frequency bands

Different configurations for up to 32 frequency bands are available.

Possibilities:

- Broadband RMS value
- Speed-dependent RMS value
- Noise



Hardware \ X20 system \ X20 modules \ Other modules \ X20CM4810 \ Register description \ Frequency bands

Broadband RMS value

This is the root mean square between the configured minimum frequency up to the configured maximum frequency.

The following signal sources are available for selection:

- Raw vibration acceleration signal
- Raw velocity signal
- Enveloped vibration acceleration signal
- Enveloped velocity signal



The harmonic frequencies of the configured frequency domain can also be calculated as well.

Speed-dependent RMS value

This is the root mean square of a movable window. The standardized switching frequency and a tolerance have to be configured to define the window. This window is coupled to a dynamic velocity.

The window results as follows:

- Minimum frequency = (speed * switching frequency) - tolerance
- Maximum frequency = (speed * switching frequency) + tolerance

The following signal sources are available for selection:

- Raw vibration acceleration signal
- Raw velocity signal
- Enveloped vibration acceleration signal
- Enveloped velocity signal



The harmonic frequencies of the window can also be included in the calculation.

Noise

This is the noise of a quadrant where the configured maximum frequency (low-pass filter for raw signal, low-pass filter for envelope signal) is divided by 4 and configured for whichever of these 4 quadrants the noise has to be determined.

The following signal sources are available for selection:

- Raw vibration acceleration signal
- Raw velocity signal
- Enveloped vibration acceleration signal
- Enveloped velocity signal



This analysis allows slippage to be effectively measured, for example. The higher the friction, the more noise that is created.

4.1.6 Analog input

The acceleration sensor input can also be used as an analog input with different special functions.

The following functions are available:

- Normal input function
- Characteristic value calculation
 - in continuous mode
 - in trigger mode (single shot)



Hardware \ X20 system \ X20 modules \ Other modules \ X20CM4810 \ Register description \ Analog input functions

Normal input function

The last 8 measured values are always averaged and the result is made available. Here the direct input signal (raw signal) is used.

Characteristic value calculation

One of the following parameters is determined:

- Mean value
- Peak value
- RMS value
- Crest factor

Available as a signal source for the parameters, there is either the raw signal filtered to the configured sampling frequency (low-pass filter for raw signal) or the raw signal filtered to 10 kHz. In addition, you can specify for how many sampled values to calculate the respective safety characteristic. The time between 2 samples depends on the configured maximum frequency (low-pass filter for raw signal).

Continuous mode

The function can be enabled via the start signal – measurement is carried out continually.

Trigger mode (single shot)

The function can be enabled via a trigger – only one measurement is carried out.

4.1.7 Functionality explained with the example of a fan

Initial situation

The condition monitoring module uses an acceleration sensor to monitor a rotating fan. Damage to the fan is causing an imbalance in its rotation.

Measurement with sensors

The mechanical imbalance is measured by an acceleration sensor and converted into an analog signal.

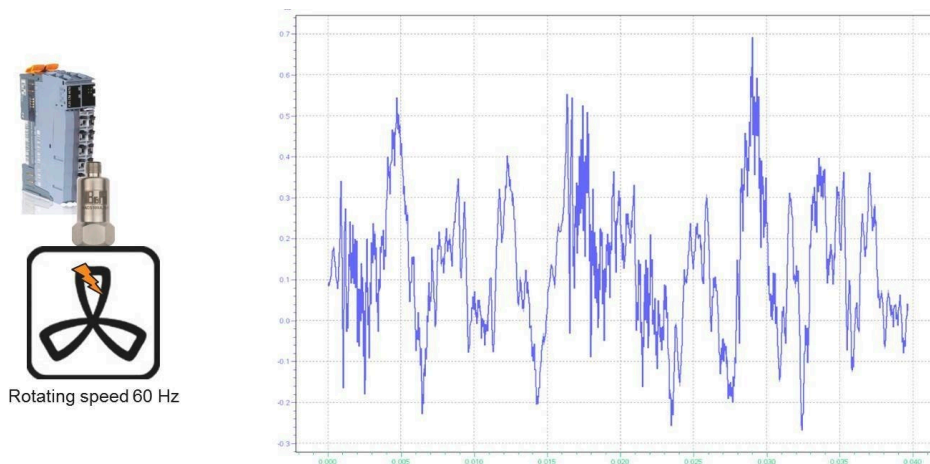


Figure 26: Raw acceleration sensor signal

Signal evaluation and calculation in the module

Using an FFT algorithm, the frequency spectrum is calculated from the raw signal. This frequency spectrum illustrates the disturbance frequencies caused by the imbalance. The module can be configured to read specific values that help determine the nature of the imbalance.

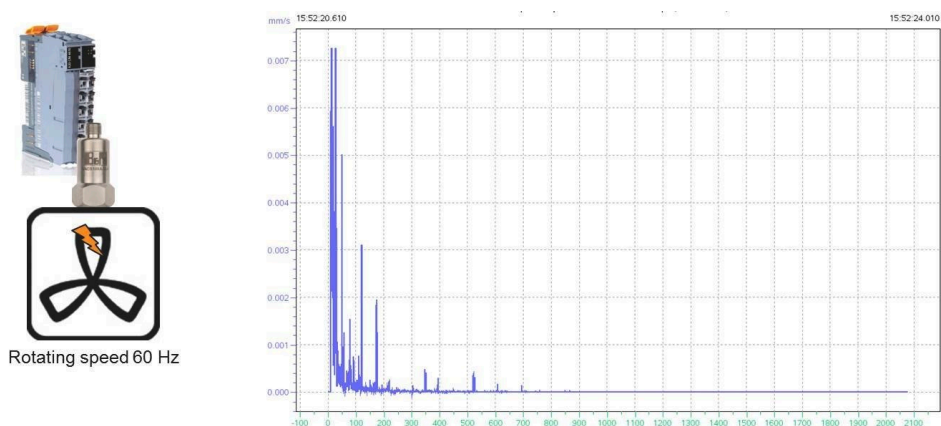


Figure 27: Frequency spectrum

Module settings in Automation Studio

In the first step, the parameters and characteristic values provided by the module can be drawn on.

For a more detailed analysis, the frequency bands can be used as help. The configured damage frequency is used to observe the frequency domain of the imbalance.



The switching frequency (excitation frequency) corresponding to the rotary frequency of the imbalanced rotor.

The measured values are output as vibration acceleration [g] or vibration velocity. They provide information about the magnitude of the imbalance. The larger the measured value, the greater the imbalance.

The measured values are compared against the machine's limit values. This is where long-term recording of data over the entire lifecycle of the machine becomes very important.

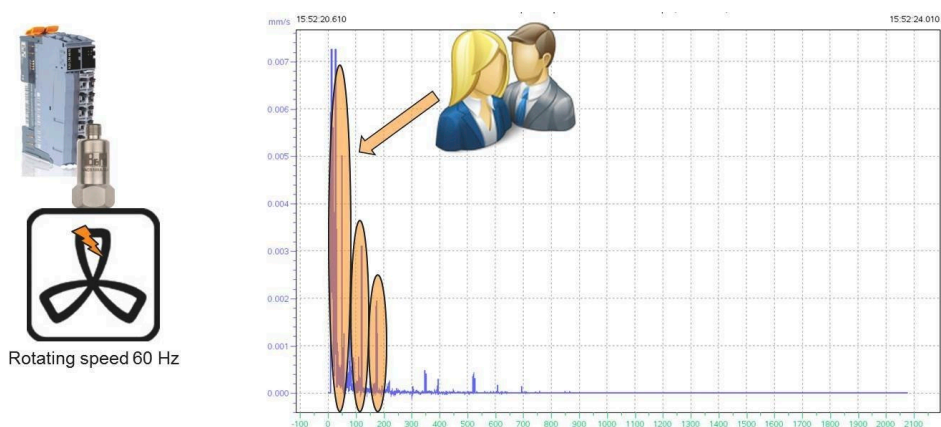


Figure 28: Frequency spectrum using frequency bands

Interpreting the results

The measured values provide insights into the condition of the machine. They can be used to determine whether there is any damage and what type of damage it is. The severity of the damage can only be judged in relation to the machine's limit values.

This is why it is so essential to monitor and record data for a machine continuously over its entire life span. With historical trend recordings, the machine can be permanently monitored to detect the signs of developing damage as quickly as possible. Broadband values provide a good overview of the general condition of the machine, and should therefore always be monitored.

It can be very helpful to set warning and alarm limits to ensure that detection is prompt and reliable.

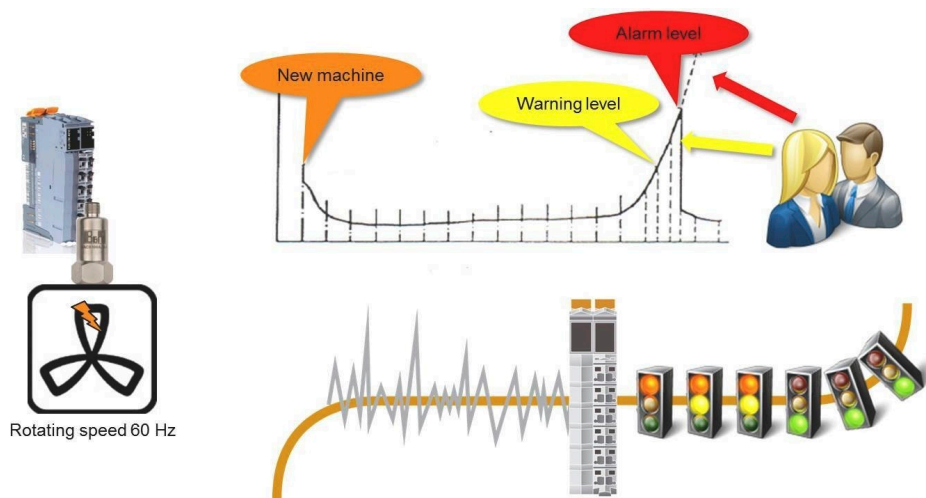


Figure 29: Interpreting the results

5 Practical applications for damage detection

Using the training hardware, some practical error scenarios can be run through and analyzed. The structure comprises one motor, two self-aligning ball bearings on the motor shaft, and a fan as the mechanical system.

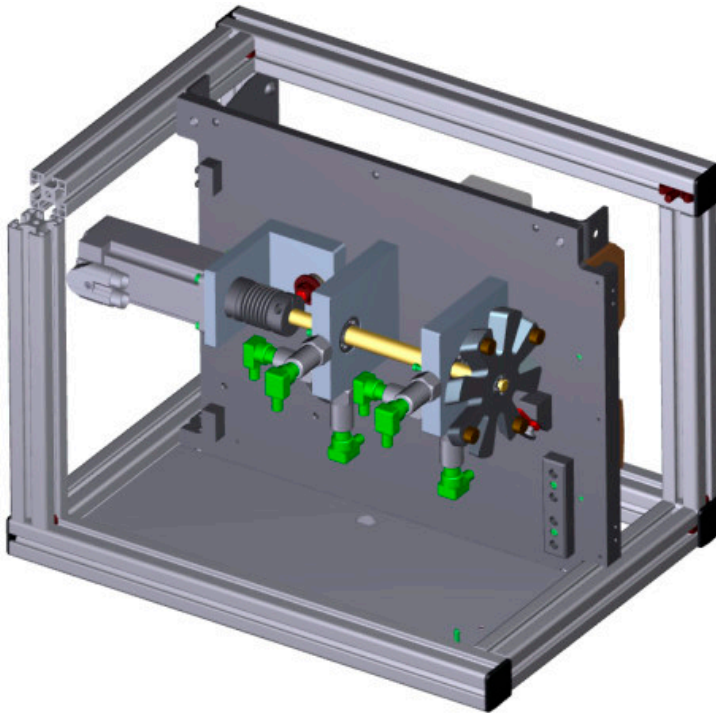


Figure 30: Structure of the training hardware

Two condition monitoring modules are used and three acceleration sensors per module are connected. The sensors nearer to the motor are connected to the 1st module (Bearing A), the sensors nearer to the fan are connected to the 2nd module (Bearing B).

Three damage scenarios can be simulated via the structure:

- Imbalance
- Periodic impact
- Bearing damage



Additional practical applications can be found in the data sheet of the module.

5.1 Analysis of imbalances

An imbalance on the mechanics can arise due to several reasons – e.g. in the case of a fan, it may be dirty or damaged. Imbalance leads to vibrations and increased wear, particularly at high speeds, which is why counterweights are applied to compensate for this as counterbalance. In practice, it is never possible to fully compensate for this, meaning each rotating body always has residual imbalance.



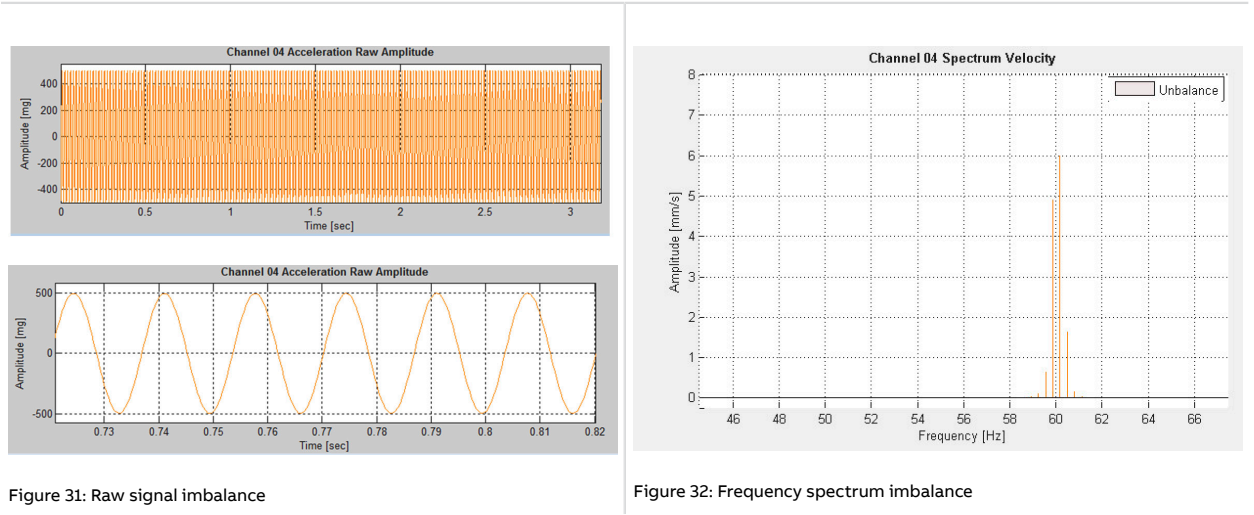
The X20CM4810 condition monitoring module can only measure the intensity of the imbalance, not its position. For this reason, it cannot be used for balancing.

A possible imbalance can be recognized via the parameters and characteristic values calculated by the module. For an exact analysis, a frequency band should be implemented where the relationship of the imbalance is visible in the frequency spectrum in the table below.

Frequency in raw signal spectrum	Frequency in envelope spectrum
1 x fn	-

Table 3: Frequency of imbalance

fn ... nominal speed



Error simulation

Magnetic weights can be attached to the blades to simulate a mechanical imbalance.

5.1.1 Exercise – Measuring an imbalance

This exercise is performed directly on the training structure. The first step is to take measurements on a balanced fan in order to obtain controlled values that can be used for comparison. The imbalance is simulated by attaching magnetic weights to the fan.

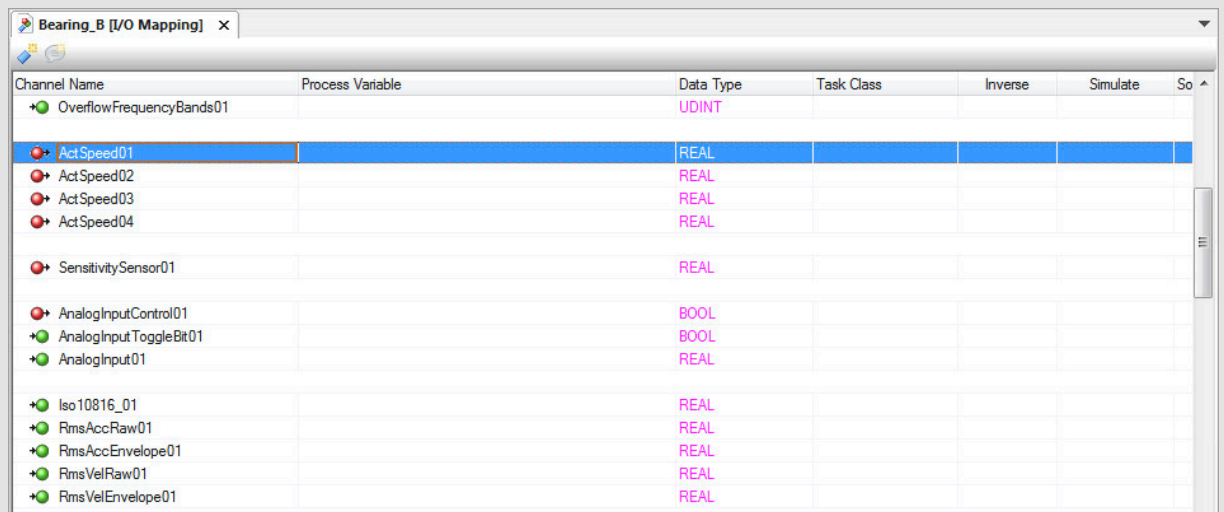
When an X20CM4810 is added, each channel (4 channels per module) is assigned 8 frequency bands. The first frequency band assigned to each channel (FrequencyBand01, 09, 17, 25) is already configured to measure imbalance.

Exercise: Measuring an imbalance

- 1) Move the motor via the motion test window or via mapp component MpAxis
 - "Controller - Switch on" in the motion test window or "Power" via the MpAxis component
 - "Homing" in the motion test window or "Home" via MpAxis component
 - "Basis Movement - Positive" in the motion test window or "Move Velocity" via the MpAxis component
- 2) The current speed of the motor in Hertz must be transferred to the module via the I/O mapping of the module.



The channel ActSpeed01 to 04 must either be forced to the current speed or a corresponding variable must be connected to the channel.



Channel Name	Process Variable	Data Type	Task Class	Inverse	Simulate	So
OverflowFrequencyBands01		UDINT				
ActSpeed01		REAL				
ActSpeed02		REAL				
ActSpeed03		REAL				
ActSpeed04		REAL				
SensitivitySensor01		REAL				
AnalogInputControl01		BOOL				
AnalogInputToggleBit01		BOOL				
AnalogInput01		REAL				
Iso10816_01		REAL				
RmsAccRaw01		REAL				
RmsAccEnvelope01		REAL				
RmsVelRaw01		REAL				
RmsVelEnvelope01		REAL				

Figure 33: Current speed of the motor - ActSpeed01 to ActSpeed04 correspond to the 4 different channels



If the axis moves e.g. at 5000 units/s and 1000 units represent a revolution, then the motor runs at 300 rpm.
This results in a current speed of 5 Hz → ActSpeed [Hz] = 5



Hardware \ X20 system \ X20 modules \ Other modules \ X20CM4810 \ Register description \ General registers \ ActSpeed

- 3) Check the frequency bands configured for imbalance (FrequencyBand01, 09, 17) and make a note of the values



Since an imbalance has not yet been simulated, the values for a balanced machine are measured here. In the real world, this is done during commissioning with the machine configured optimally. These results are then used for comparison when measuring imbalances later on.

- 4) Stop motor to remeasure with imbalance
- 5) Attach magnetic weight to fan in order to simulate an imbalance
- 6) Start motor and set ActSpeed → Repeat measurement
- 7) Compare results against values from balanced machine and interpret
- 8) Increase speed of motor → Adjust parameter for ActSpeed [Hz]
- 9) Repeat imbalance measurement - Compare the values and interpret the results



The higher the speed, the greater the possibility of imbalance.

5.2 Analyzing periodic impacts

There are several reasons why an impact on the mechanics can arise, e.g. in the case of a fan, defective mechanical parts or foreign bodies can cause a periodic impact by touching the fan.

In addition, components strike their counterparts on each revolution. This in turn causes the attachment parts to vibrate at their natural frequency. Envelope analysis can be used to separate the causes of impact.



It also looks very similar to individual parts striking each other when parts in the machine are loose. These two causes of damage cannot be analyzed separately.

Frequency in raw signal spectrum	Frequency in envelope spectrum
-	1 x fn

Table 4: Frequency of a periodic impact

fn ... nominal speed

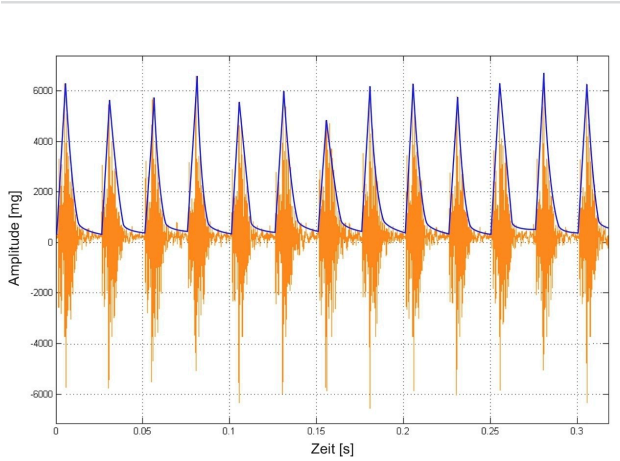


Figure 34: Raw periodic impact signal

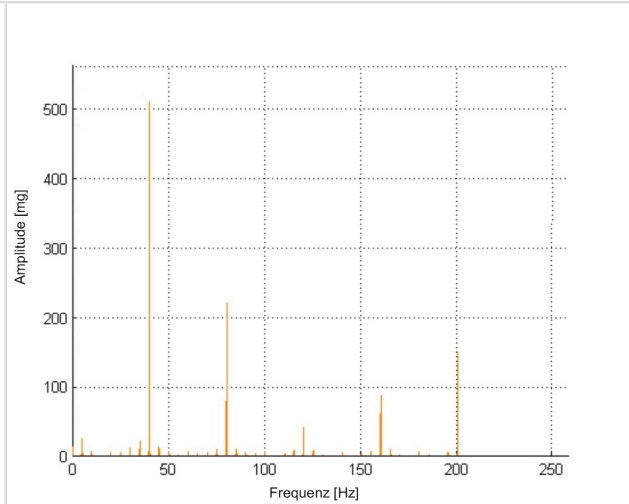


Figure 35: Frequency spectrum periodic impact

Error simulation

A small plate is mounted so that it brushes the fan blades in order to simulate a periodic impact on the mechanical structure.

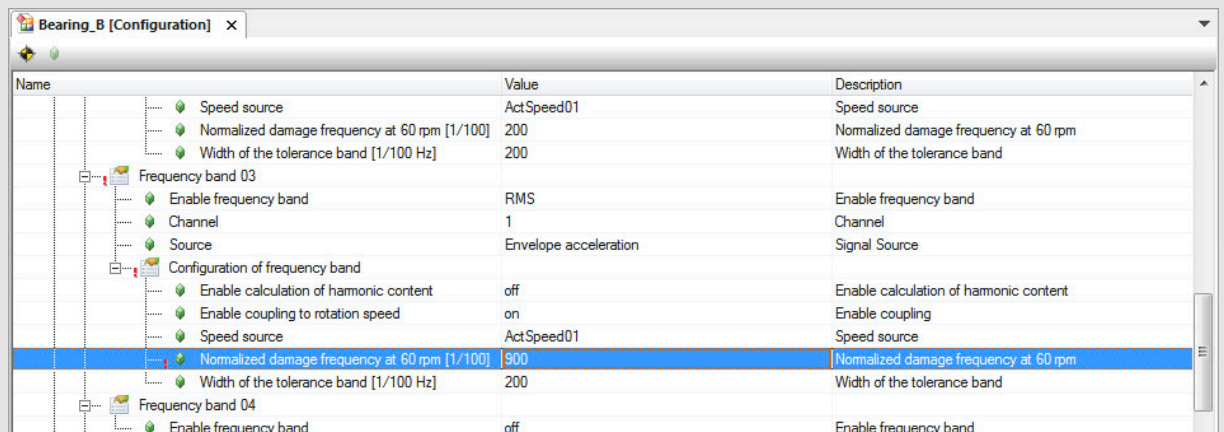
5.2.1 Exercise – Measuring a periodic impact

This exercise is performed directly on the training structure. The first step is to take measurements on a fan without any impact events in order to obtain controlled values that can be used for comparison. The periodic impact is simulated using a small metal or plastic plate mounted to the right of the fan so that it comes in contact with the rotor blades.

Frequency bands need to be configured for measuring a periodic impact.

Exercise: Measuring a periodic impact

- 1) Configure frequency bands 03, 11, 19 via the module configuration for measuring a periodic impact



Name	Value	Description
Speed source	ActSpeed01	Speed source
Normalized damage frequency at 60 rpm [1/100]	200	Normalized damage frequency at 60 rpm
Width of the tolerance band [1/100 Hz]	200	Width of the tolerance band
Frequency band 03		
Enable frequency band	RMS	Enable frequency band
Channel	1	Channel
Source	Envelope acceleration	Signal Source
Configuration of frequency band		
Enable calculation of harmonic content	off	Enable calculation of harmonic content
Enable coupling to rotation speed	on	Enable coupling
Speed source	ActSpeed01	Speed source
Normalized damage frequency at 60 rpm [1/100]	900	Normalized damage frequency at 60 rpm
Width of the tolerance band [1/100 Hz]	200	Width of the tolerance band
Frequency band 04		
Enable frequency band	off	Enable frequency band

Figure 36: Configuring frequency bands



The fan possesses 9 rotary blades, which results in 9 impacts per revolution.

The value of "Normalized damage frequency at 60 rpm" must be adjusted here → Value to be set = 900 (where the unit is specified as [1/100])

- 2) Move the motor via the motion test window or mapp component MpAxis (see exercise "Measuring an imbalance")
- 3) Current speed of the motor in Hertz must be transferred to the module again (see exercise "Measuring an imbalance")
- 4) Check the frequency bands configured for periodic impact FrequencyBand03, 11, 19 and make a note of the values → A value between 1 - 3 mg should be provided
- 5) Attach plate to ETA and repeat measurement with impact



Don't set the speed (motor speed) too high, because this could produce too much noise.

- 6) Renewed consideration of the values of the frequency bands
 - ° Interpreting the results
 - ° Measurement with condition parameters → Peak, Crest Factor Raw, Kurtosis Raw

Additional exercise: Measuring using analog input functionality

- 1) Switch on analog-input functionality of the channels via the module configuration → Trigger mode

Bearing_B [Configuration]		
Name	Value	Description
Bearing_B		
Function model	default	Module operating mode
General		
Module configuration		
High frequency configuration	1000 Hz	High frequency configuration for envelope and HF parameters
Autogain delay [measure cycle]	50	Autogain increase only if x measure cycles without overflow
Channel 01		
IEPE current supply	on	IEPE current supply on/off
Maximum frequency raw	2000 Hz	Upper cut-off frequency of raw signal
Maximum frequency envelope	2000 Hz	Upper cut-off frequency of envelope
Enable calculation of velocity signal	off	Enable calculation of velocity signal
Scaling of AnalogInput01	128	Maximum value in both directions of AnalogInput01 [g] or [1](CF
Enable parameter calculation of AnalogInput01	on	off = AnalogInput01 shows actual input value, on = AnalogInput
Signal source for AnalogInput01	Raw signal filtered at maximum frequency raw	Signal source for parameter calculation of AnalogInput01
Source for samples of AnalogInput01	Configuration	Sample source for parameter calculation of AnalogInput01
Samples of AnalogInput01	1000	Samples for parameter calculation of AnalogInput01
Mode of AnalogInput01	Single calculation with trigger	Mode for parameter calculation of AnalogInput01
Parameter of AnalogInput01	Mean value	Calculated parameter of AnalogInput01
Channel 02		

- 2) Move the motor via the motion test window or mapp component MpAxis (see exercise "Measuring an imbalance")
- 3) Start recording with Trigger, perform measurement without plate → AnalogInput0x should return a low value
- 4) Attach plate to ETA and repeat measurement with impact



Don't set the speed (motor speed) too high, because this could produce too much noise.

- 5) Renewed consideration of the values
- 6) Consideration with different configurations
 - Which parameter calculation produced the most helpful results?
 - What effect does configuring the samples have?

5.3 Analysis of bearing damage

Bearing damage also causes a periodic impact signal. Damage to the inner or outer bearing ring or the balls themselves also causes a periodic impact during rotation.

Many types of bearing damage are caused by imprecisions in the bearing surface such as material damage or micro-cracks. These pits are rolled over by the roller elements and cause impacts on the roller bearing and its attachment parts.



The mechanism is very similar to the striking of a bell: The clapper strikes the body of the bell, and the bell starts vibrating at its natural frequency.

In the case of the bearing, each time the roller moves over the damaged area, it is like striking the clapper and the roller parts and attached parts start to vibrate.

Outer ring damage

In most cases the outer ring remains stationary while the inner ring turns. This gives a clearly defined fixed load zone. Most damage occurs in this load zone. If pitting or other surface damage occurs, it is rolled over by the rolling elements. Vibrations occur when rolling over, which can be measured on parts of the housing.

Frequency in raw signal spectrum	Frequency in envelope spectrum
-	1 x fa, 2 x fa, 3 x fa ...

Table 5: Frequency of outer ring damage

fa ... frequency of the outer ring damage

Inner ring damage

Any inner ring damage that occurs travels with the rotating shaft. Due to the different rotary speeds of the revolving roller elements and the inner ring, pronounced modulations occur. As a result, inner ring damage frequencies are usually shown with sidebands in the spectrum.

Frequency in raw signal spectrum	Frequency in envelope spectrum
-	$i \times f_i \pm f_n$

Table 6: Frequency of inner ring damage

fi ... frequency of the inner ring damage
fn ... nominal speed



Details of bearing damage frequencies are normally provided by the manufacturer and can be taken from the data sheets for the bearings.

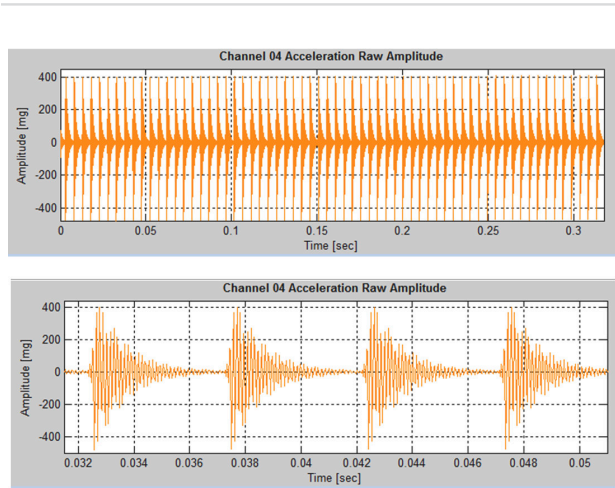


Figure 37: Raw signal – Bearing damage

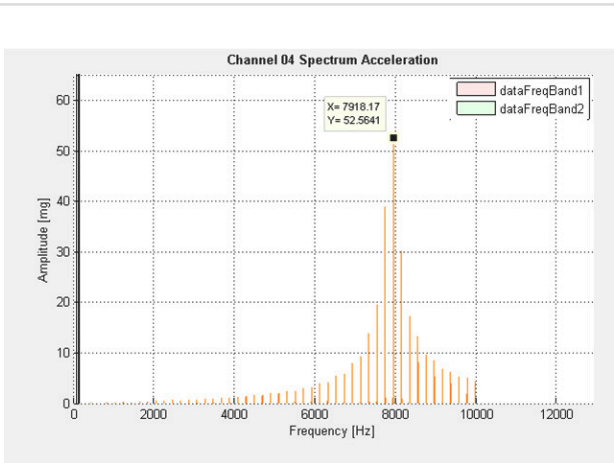


Figure 38: Raw signal – Frequency spectrum

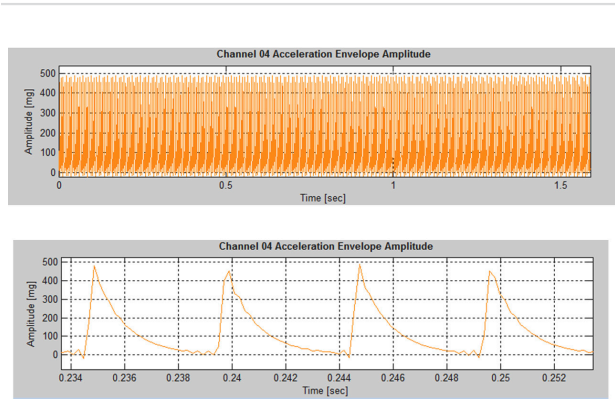


Figure 39: Envelope signal – Bearing damage

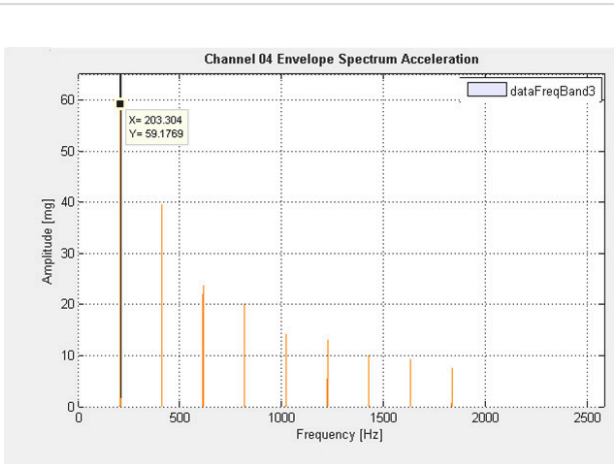


Figure 40: Envelope signal – Frequency spectrum

Error simulation

One of the two self-aligning ball bearings is defective. The data sheet of the manufacturer can be used for a detailed analysis.

5.3.1 Exercise – Measuring bearing damage

Frequency bands need to be configured to measuring bearing damage.

Exercise: Measuring bearing damage

1) There are three relevant values

- Damage frequency on inner ring
- Damage frequency on outer ring
- Damage frequency directly on the ball



All three 3 potential types of damage need to be monitored → It's best to configure a separate frequency band for each value.

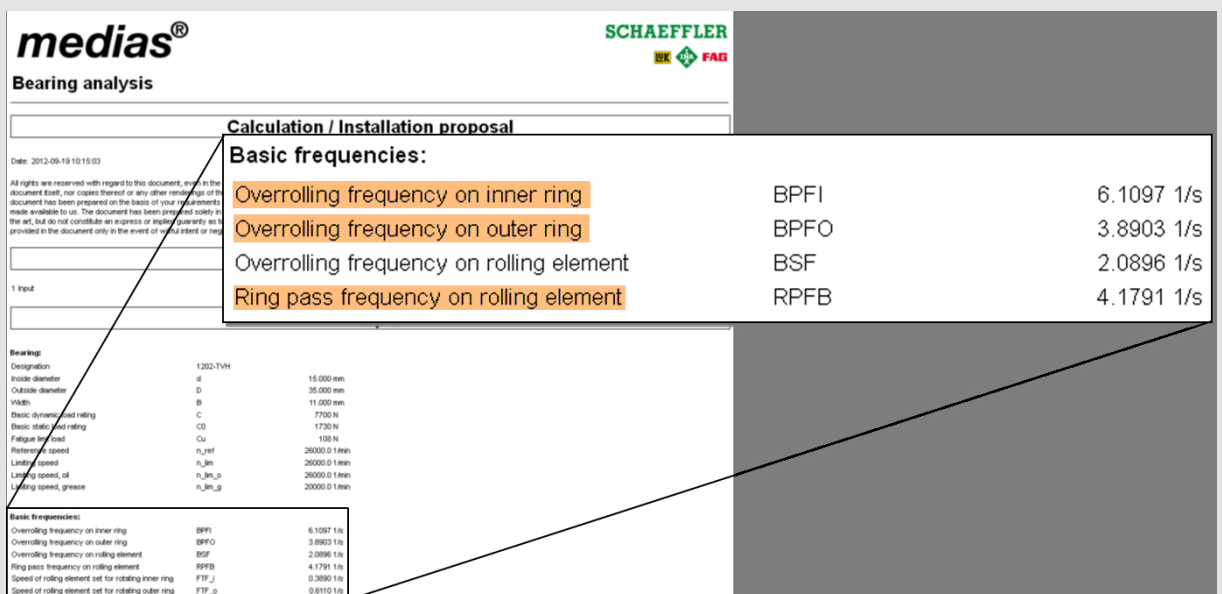


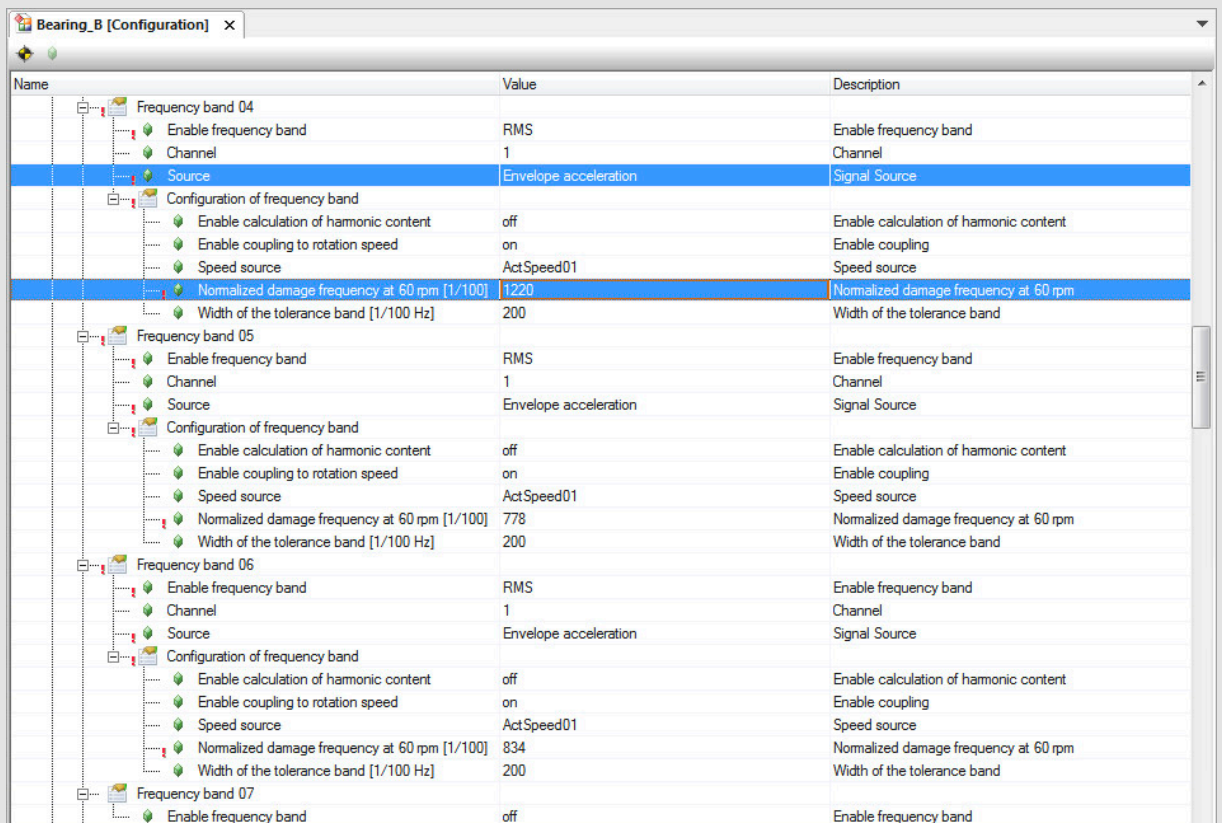
Figure 41: Bearing data sheet from the manufacturer



Special case: Self-aligning ball bearings

Since the bearings are self-aligning ball bearings, when there is damage to the bearing that generates an impact on both passing balls you need to select **2x the specified damage frequency**.

- #### 2) Configure three frequency bands per channel (e.g. 04 - 06, etc.) via the module configuration for measuring a bearing damage signal.



Name	Value	Description
Frequency band 04		
Enable frequency band	RMS	Enable frequency band
Channel	1	Channel
Source	Envelope acceleration	Signal Source
Configuration of frequency band		
Enable calculation of harmonic content	off	Enable calculation of harmonic content
Enable coupling to rotation speed	on	Enable coupling
Speed source	ActSpeed01	Speed source
Normalized damage frequency at 60 rpm [1/100]	1220	Normalized damage frequency at 60 rpm
Width of the tolerance band [1/100 Hz]	200	Width of the tolerance band
Frequency band 05		
Enable frequency band	RMS	Enable frequency band
Channel	1	Channel
Source	Envelope acceleration	Signal Source
Configuration of frequency band		
Enable calculation of harmonic content	off	Enable calculation of harmonic content
Enable coupling to rotation speed	on	Enable coupling
Speed source	ActSpeed01	Speed source
Normalized damage frequency at 60 rpm [1/100]	778	Normalized damage frequency at 60 rpm
Width of the tolerance band [1/100 Hz]	200	Width of the tolerance band
Frequency band 06		
Enable frequency band	RMS	Enable frequency band
Channel	1	Channel
Source	Envelope acceleration	Signal Source
Configuration of frequency band		
Enable calculation of harmonic content	off	Enable calculation of harmonic content
Enable coupling to rotation speed	on	Enable coupling
Speed source	ActSpeed01	Speed source
Normalized damage frequency at 60 rpm [1/100]	834	Normalized damage frequency at 60 rpm
Width of the tolerance band [1/100 Hz]	200	Width of the tolerance band
Frequency band 07		
Enable frequency band	off	Enable frequency band

Figure 42: Configuring frequency bands



The value of "Normalized damage frequency at 60 rpm" must be adjusted here. [1/100] is again specified as the unit.

- 3) Move the motor via the motion test window or mapp component MpAxis (see instructions under "Workshop 2: Measuring a mechanical imbalance")
- 4) Current speed of the motor in Hertz must be transferred to the module again (see "Workshop 2: Measuring a mechanical imbalance")
- 5) Find out:
 - How does increased speed affect the results?
 - Which bearing is defective → Bearing A or Bearing B?
 - Where is the damage on the defective bearing → On the inner ring, outer ring or directly on the ball?

6 Condition monitoring in a real machine environment

Basically three central tasks can be identified for the implementation of condition monitoring in the real machine environment.

- Collecting data
- Recording
- Reporting

mapp Technology is used to implement individual tasks since many useful basic functions are present there.



Figure 43: mapp Technology

A condition monitoring solution can be a fixed component of a series-produced machine. However, a solution can also be developed as a standalone product, which is subsequently retrofitted to existing systems or just sold separately.



The measured values collected can deviate from machine to machine. Detailed knowledge about the machine is necessary to be able to make a classification in relation to the limit value or alarm, e.g. via the mechanical design. During vibration monitoring, for example, this is the only way to define which vibration values are critical for the machine.



Mathematical procedures, e.g. the standard deviation, for analyzing the measured values collected can be used. Deviations can be determined this way, but it is not necessarily possible to relate this to the state of the machine.

If possible, a comparison of good and bad test cases should be carried out.



If vibration information should serve as the basis for determining the state of a machine, then, to begin, the raw data should always be analyzed too - please see [of the raw data](#).

By doing so, basic information and interesting frequency domains can be determined and the configuration of the module adjusted correspondingly – e.g. recording duration, frequency resolution of the FFT, frequency bands, etc.

6.1 Collecting data

To begin, relevant information/physical quantities must be collected. Corresponding sensors are built into the machine for this.

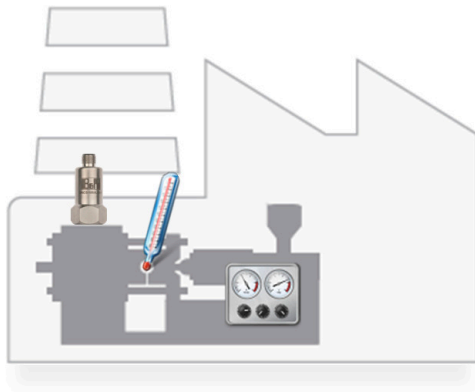


Figure 44: Collecting information

By connecting sensors to the associated I/O modules, information is treated for post-processing. Depending on the range of functions of the I/O module, certain analysis options are available directly via the module.



If, for example, the imbalance of a machine should be monitored, then a vibration sensor is used. In combination with the condition monitoring module, the imbalance at the current speed can already be determined purely via the configuration of the frequency bands.

Exercise: Detecting an imbalance

- 1) Configuration of the three channels of the two condition monitoring modules for imbalance
- 2) Transmit the current speed of the motor to the modules

6.2 Recording

In addition to collecting information, the information should also be saved for future traceability. In some circumstances, a relationship between different events can be created on the machine.

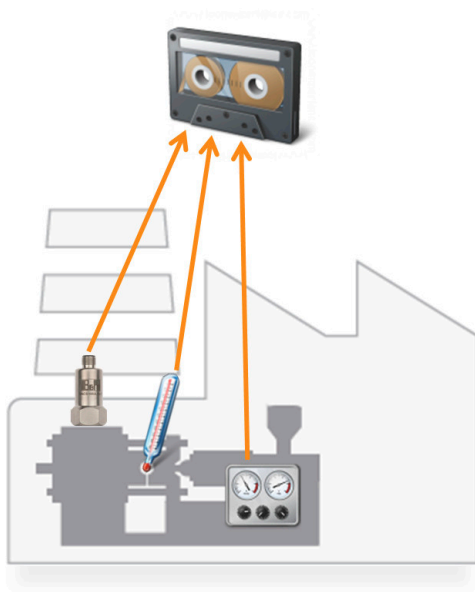


Figure 45: Recording information

Information can be logged quickly and easily in a file via the data logging functionality (MpData).

Several variables for storage can be registered and the save interval defined. In addition, there are examples for implementation in the HMI application.

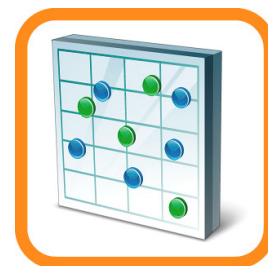


Figure 46: Data logging



Additional information and a detailed description can be found in the help documentation for MpData.

Exercise: Recording an imbalance

- 1) Connect frequency bands to variables via the I/O mapping
- 2) Record the frequency bands for monitoring the imbalance using MpData (CSV file)
- 3) Recording duration 1 sec
- 4) Vary the speed of the motor

Exercise: Analysis of the recorded data

- 1) Recorded data can be uploaded from the controller via FTP.
- 2) Open file with Excel and generate diagram

6.3 Reporting

The collected information should also be monitored for compliance with limits. If there is a violation of a limit, then this should be reported and also logged.

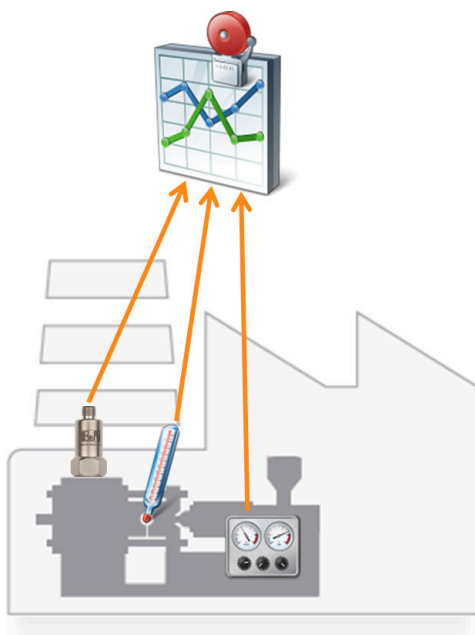


Figure 47: Reporting the status

Messages for limit violations can be triggered via the alarm system (MpAlarmX).

Different properties and language-dependent texts can be defined for the messages.

In addition, there are examples for implementation in the HMI application.



Figure 48: Alarm system



Additional information and a detailed description can be found in the help documentation for MpAlarmX.



Additional exercise: Monitor limits and trigger messages

The imbalance values should be monitored for compliance with limit values (e.g. standard deviation, mean value, etc.) and a message should be triggered for any violations.

Messages are not only triggered or displayed locally but can also be sent via the messaging system (MpTweet) in the form of a text message.

A notification hierarchy and language-dependent texts can be defined.

A message can be acknowledged via response message.



Figure 49: Messaging system



Additional information and a detailed description can be found in the help documentation for MpTweet.

7 APROL ConMon

APROL ConMon solution is based on the APROL process control system. With it, B&R offers a solution for measuring, recording and evaluating all relevant condition parameters to provide optimal support for the continual improvement process. APROL ConMon can be used as a standalone solution or integrated into existing APROL process control systems. It can be customized to monitor a single machine, an entire factory, a building, a process or a plant.



Figure 50: APROL ConMon

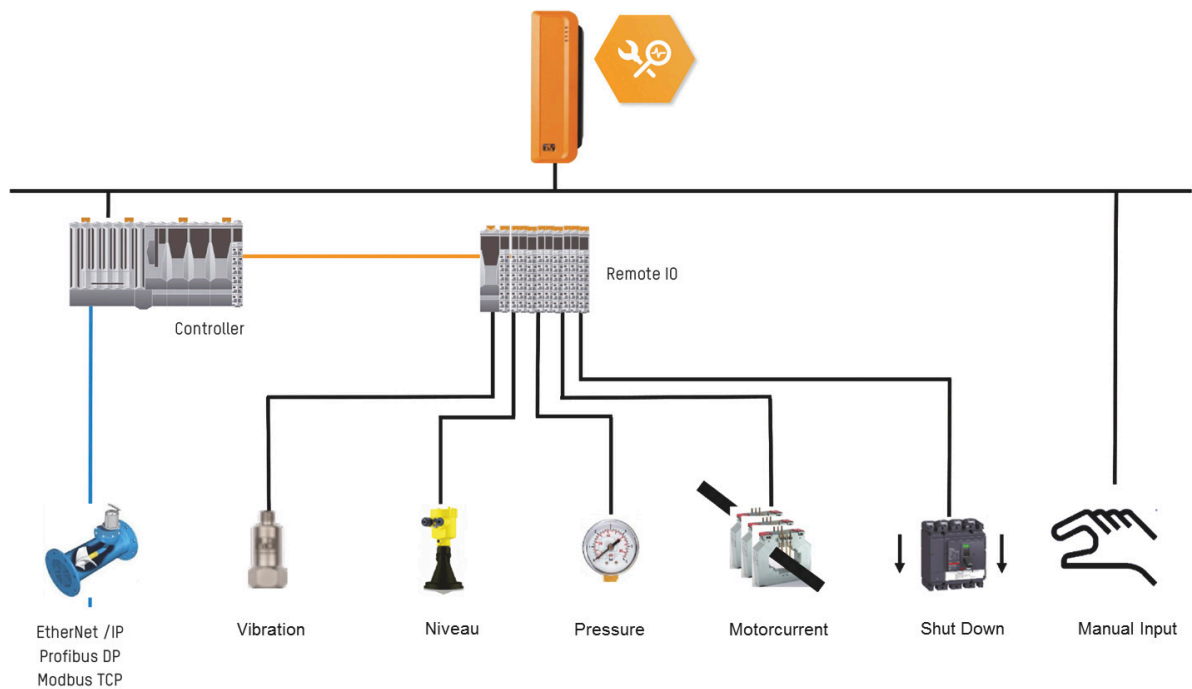


Figure 51: APROL ConMon system structure

Condition parameters are collected using different I/O modules. A comprehensive library makes powerful control modules available.

Numerous ready-made reports are available in APROL. Separate reports can be created and interactive dashboards shown.

8 Summary

Condition monitoring involves constantly monitoring the health of a machine over years of operation. The availability, maintenance and service life of a machine can be considerably improved and optimized through the application of a condition monitoring solution.

There is a wide variety of physical parameters that, when monitored constantly, can provide key insight into the condition of a machine and the efficiency with which it is operating. B&R's condition monitoring modules make it possible to systematically monitor and analyze the health and efficiency of a machine using vibration measurement.

The module can be used to identify damage and potential sources of error such as imbalance, misalignment, bearing damage and periodic impact, all of which generate measurable vibrations throughout the mechanical structure of the machine.



Figure 52: Condition monitoring

This training module provides information about the use of condition monitoring in all types of industries. Signal evaluation based on vibration analysis is described in detail and applied in the course of various workshops in order to identify imbalance, periodic impacts and bearing damage.

9 Appendix

9.1 Uploading raw data

It's possible to upload the raw data saved in a buffer on the module using the AsIOVib library. Either the current measurement (8k) or the last eight measurements (64k) can be uploaded.



Programming \ Libraries \ Direct I/O access \ AsIOVib

Information can be quickly and easily saved to a document via the recipe functionality (MpRecipe).

Several variables for storage can be registered and the file type (CSV or XML) defined.

In addition, there are examples for implementation in the HMI application.



Figure 53: Saving the data



Additional information and a detailed description can be found in the help documentation for MpRecipe.



Additional exercise: Upload and save raw data

The buffer of the raw signal (X value, Y value) of the acceleration must be uploaded and saved via MpRecipe in a CSV file.

The data can be shown in Excel (data still has to be rearranged) or processed further.

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